

TOPOLOGICAL PHOTONIC COMPUTING: INDEPENDENT EXPERIMENTAL CORROBORATION FROM BUCHER ET AL. (NATURE, 2026)

*A Technical Note on Convergent Findings Between Theoretical Framework and Experimental
Observation*

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ABSTRACT

In January 2026, we published a theoretical framework for photonic computing based on information storage in topological dark structures—phase singularities and optical vortices—rather than in light propagation itself (Vanhorn, 2026; Zenodo DOI: [10.5281/zenodo.18226545](https://doi.org/10.5281/zenodo.18226545)). The framework proposed that these topological features possess quantized, conserved charge suitable for robust information encoding, and that confining media would stabilize and enhance singularity behavior.

In March 2026, Bucher et al. published in *Nature* the first direct experimental measurement of optical phase singularity dynamics at femtosecond temporal and nanometer spatial resolution. Their findings—topological charge conservation in pair creation/annihilation events, confirmation that singularities are non-particulate topological features unconstrained by the speed of light, and amplification of singularity dynamics in slow-light confining media—are consistent with several foundational claims of our theoretical framework.

This Technical Note formally documents the points of convergence between the two independent bodies of work, identifies new implications for the framework’s engineering pathway, and delineates the specific claims that remain experimentally unvalidated.

1. INTRODUCTION

The Topological Photonic Computing (TPC) framework (Vanhorn, 2026) proposed a paradigm shift in photonic information processing: rather than confining photons—which intrinsically propagate—information should be encoded in their topological *absence*. Phase singularities, points where optical phase is undefined and intensity is exactly zero, carry quantized topological charge that is conserved, discrete, and robust against continuous perturbation. The framework formalized this “Negative Form Principle” through four axioms, developed an information-theoretic architecture based on optical vortex topology, and proposed a five-phase experimental validation pathway.

The framework was published on Zenodo in January 2026, prior to and independently of the experimental work discussed here.

In March 2026, Bucher et al. published in *Nature* a study titled “Superluminal correlations in ensembles of optical phase singularities.” Working with hexagonal boron nitride (hBN) membranes supporting hyperbolic phonon polaritons, they achieved the first direct measurement of phase-space correlations in singularity ensembles, resolving individual singularity dynamics at ~ 20 nm spatial and ~ 3 fs temporal resolution. Their study was motivated by fundamental physics—extending the understanding of topological defect universality across wave systems—with no reference to information storage or computing applications.

The convergence between a theoretical computing framework and an independent experimental physics study, conducted from different premises and arriving at compatible conclusions about the same physical objects, warrants formal documentation. This Technical Note provides that record.

Disclosure: The author of this note is the author of the TPC framework. This analysis represents the framework author’s assessment of how the new experimental findings relate to the previously published theoretical claims. The experimental work of Bucher et al. was conducted entirely independently.

2. SUMMARY OF RELEVANT THEORETICAL CLAIMS

The TPC framework makes several claims about the physical properties of optical phase singularities that are relevant to the Bucher et al. findings. We restrict this summary to those specific claims, referring the reader to the full framework (Vanhorn, 2026) for the complete theoretical development.

2.1 Topological Charge as Information Carrier (Axiom 1, Sections 1.2, 7.2)

The framework’s first axiom—the Negative Form Principle—asserts that information in photonic systems is stored in the topology of absence: stable voids defined by boundary conditions. The topological charge l of a phase singularity, computed as:

$$l = (1/2\pi) \oint \nabla \varphi \cdot d\mathbf{l}$$

is quantized (integer-valued), conserved under continuous field evolution, and robust against perturbation. These properties—quantization, conservation, and robustness—are the three requirements for a viable information carrier.

2.2 Phase Singularities as Non-Particulate Topology (Section 1.2, Chapter 2)

The framework explicitly treats singularities as topological features of a field, not as particles or quasi-particles. The Negative Form Principle draws on Goethean phenomenology to characterize darkness as a “positive polar principle” with structure, distinct from both the presence of light and from particulate matter. The framework’s information-theoretic architecture depends on this distinction: the data carrier is topology, not substance.

2.3 Confining Media Stabilize Singularity Structures (Axioms 2–3, Chapters 4, 9)

The framework proposes that engineered confining media—specifically cholesteric liquid crystals—would stabilize phase singularities for information storage. Axiom 2 (the Phase Transition Principle) requires confinement as one of four conditions for transitioning from propagating light to stored topology. The engineering framework (Chapter 9) specifies material parameters for a confining substrate designed to support stable vortex structures.

2.4 Singularity Pair Dynamics Are Relevant to Read/Write Operations (Section 8.1)

The framework’s write mechanism involves the creation of topological structures (singularity generation), and erasure involves their removal (singularity annihilation). The dynamics of these creation and annihilation events are implicitly central to the framework’s engineering specifications, though the original document did not develop detailed predictions about the velocity dynamics near these transitions.

3. THE BUCHER ET AL. EXPERIMENTAL FINDINGS

Bucher et al. (2026) achieved the first direct measurement of phase-space correlations—joint position and velocity distributions—in ensembles of optical phase singularities. We summarize the findings relevant to the TPC framework.

3.1 Experimental Platform

The study used hexagonal boron nitride (hBN) membranes supporting hyperbolic phonon polaritons—light-matter hybrid excitations that propagate at velocities significantly below the free-space speed of light. Ultrafast transmission electron microscopy (UTEM) provided simultaneous spatial (~ 20 nm) and temporal (~ 3 fs) resolution, both exceeding the polaritonic wavelength and cycle period by an order of magnitude.

3.2 Topological Charge Conservation

The study tracked vortex-antivortex pair creation and annihilation events directly. Singularities were observed to appear and disappear exclusively in charge-conjugate pairs, preserving net topological charge. No isolated creation or destruction of individual singularities was observed. This is consistent with the topological conservation law expressed by the winding number integral.

3.3 Superluminal Velocities and Non-Particulate Character

Phase singularities were observed to accelerate to velocities exceeding the speed of light in the moments preceding pair annihilation, with velocities formally diverging at the annihilation point. Twenty-nine percent of tracked dark points exceeded c in the hBN platform—approximately 70 times the rate expected in free space. The authors emphasize that this violates no physical law: singularities carry no mass, energy, or information in transit, and their superluminal motion is analogous to the sweep of a shadow or a projected dot.

This observation confirms that phase singularities are fundamentally non-particulate. They are topological features of the field, unconstrained by the dynamical limits that govern matter and energy.

3.4 Amplification by Slow-Light Media

The superluminal behavior was paradoxically *amplified* by the slow group velocity of phonon polaritons in hBN. The confining medium did not merely preserve singularity properties but enhanced their topological dynamics. The authors attribute this to the tight confinement and slow propagation compressing the phase-space in which singularities evolve.

3.5 Liquid-Like Spatial Correlations

Singularity ensembles exhibited distance correlations resembling particles in a liquid—characteristic spatial ordering with well-defined nearest-neighbor distributions. This implies that singularities in confined media self-organize into structured ensembles with quantifiable inter-singularity spacing.

4. POINTS OF CONVERGENCE

We now map specific theoretical claims from the TPC framework onto specific experimental results from Bucher et al. For each point, we state the original claim, the experimental finding, and the nature of the convergence.

4.1 Topological Charge Conservation

Framework claim (Axiom 1, Section 7.2): Topological charge l is quantized, conserved under continuous field evolution, and robust against perturbation. These properties make it suitable as an information carrier.

Experimental finding (Bucher et al., 2026): Singularities appear and disappear exclusively in charge-conjugate pairs, conserving net topological charge across all observed creation and annihilation events.

Convergence: The experimental observations are directly consistent with the conservation law the framework invokes as the basis for information integrity. While topological charge conservation is well-established theoretically (Berry, 1978), the Bucher et al. study provides the first *direct dynamical observation* at the resolution required to track individual events. This moves the framework’s foundational claim from theoretical inference to experimental corroboration.

4.2 Non-Particulate Nature of Phase Singularities

Framework claim (Axiom 1, Chapter 2): Phase singularities are topological features of structured absence, not particles. The Negative Form Principle treats darkness as a “positive polar principle” carrying information through topology, not through substance.

Experimental finding (Bucher et al., 2026): Singularities violate the speed-of-light constraint, reaching superluminal velocities. The particle-singularity analogy breaks down. Singularities are explicitly not bound by the dynamical limits governing particles.

Convergence: The experimental confirmation that singularities are non-particulate directly supports the framework’s foundational ontological claim. The TPC framework depends on singularities being topological features rather than quasi-particles—their information-carrying capacity derives from topology (winding number), not from any particle-like property (mass, charge, energy). The Bucher et al. observation that singularities can exceed c is the strongest possible experimental evidence that these objects occupy a fundamentally different category from particles.

This point of convergence is particularly notable because the two works arrive at the same conclusion from entirely different starting points: the TPC framework from Goethean phenomenology and RS2 dimensional algebra, Bucher et al. from ultrafast electron microscopy of polariton fields.

4.3 Confining Media Enhance Singularity Behavior

Framework claim (Axioms 2–3, Chapter 9): Confining media (proposed: cholesteric liquid crystals) would stabilize phase singularities and support their use as information carriers. The Phase Transition Principle requires confinement as a condition for converting propagating light into stored topology.

Experimental finding (Bucher et al., 2026): The hBN slow-light platform amplified superluminal behavior by a factor of ~ 70 relative to free-space expectations. Confinement did not merely preserve but *enhanced* singularity dynamics.

Convergence: While the specific materials differ (hBN phonon polaritons vs. cholesteric liquid crystals), both represent engineered slow-light confining media. The experimental demonstration that confinement amplifies rather than suppresses topological behavior strengthens the framework’s engineering premise. The amplification factor ($\sim 70\times$) suggests that the choice of confining medium may offer a significant design parameter for tuning singularity properties—a variable not anticipated in the original framework.

Important caveat: The framework proposes confinement for *stabilization* (persistence over time), while Bucher et al. demonstrate confinement effects on *dynamics* (velocity amplification). These are related but distinct phenomena. The experimental work does not directly test storage stability.

4.4 Singularity Ensemble Structure

Framework claim (Section 8.1, Chapter 9): The engineering framework proposes arrays of vortex storage sites (Phase 2 validation: 10×10 minimum grid) with defined spacing and independent addressability.

Experimental finding (Bucher et al., 2026): Singularity ensembles in confined media exhibit liquid-like spatial correlations with well-defined nearest-neighbor distributions.

Convergence: The observation that singularities self-organize with characteristic spacing is directly relevant to storage array design. The inter-singularity correlations measured by Bucher et al. may set fundamental bounds on minimum storage site spacing. If singularities interact at distances comparable to proposed array pitch, the engineering framework will need to account for these correlations—either as a constraint (minimum spacing) or as a resource (correlated states for error detection).

5. NEW IMPLICATIONS FOR THE FRAMEWORK

The Bucher et al. findings suggest several refinements and extensions to the TPC framework that were not part of the original formulation.

5.1 Velocity Divergence as Write/Erase Signature

The observation that singularity velocities diverge near pair annihilation implies a characteristic field-gradient signature at the moment of topological transition. For the framework’s write/erase cycle, this suggests:

- **Write confirmation:** The creation of a singularity pair may produce a detectable velocity/gradient transient that serves as a natural write-verification signal, potentially eliminating the need for a separate read-after-write step.
- **Erase dynamics:** Annihilation involves extreme local field gradients. The energy dissipated during these events should be characterized, as it may set bounds on minimum erase energy and maximum erase speed.
- **Transition timescale:** The femtosecond dynamics observed by Bucher et al. suggest that the topological transitions themselves are extremely fast—potentially sub-picosecond—even if the overall write cycle is limited by slower processes (condensation, alignment).

5.2 hBN as Alternative or Complementary Substrate

The hBN platform used by Bucher et al. offers properties distinct from the cholesteric liquid crystals proposed in the TPC framework:

Property	Cholesteric LC (proposed)	hBN Polaritons (Bucher et al.)
Confinement mechanism	Helical geometry	Hyperbolic dispersion
Operating wavelength	Visible–NIR	Mid-infrared (phonon)
Reconfigurability	High (electric field tuning)	Low (fixed crystal)
Fabrication	Established LC methods	Exfoliation/growth
Singularity dynamics	Not yet measured	Characterized at fs/nm scale

These platforms may be complementary: hBN for characterization and fundamental studies, LC for reconfigurable storage devices. The framework’s experimental validation pathway (Chapter 12) should consider incorporating hBN-based measurements alongside LC experiments, particularly for Phase 1 (vortex stability characterization).

5.3 Inter-Singularity Correlations and Storage Density

The liquid-like correlations observed by Bucher et al. introduce a previously unconsidered variable into the framework’s storage density projections (10^{15} – 10^{19} bits/cm³, Chapter 11). If singularities in confined media maintain characteristic nearest-neighbor distances, the achievable density will be bounded by these correlation lengths. Conversely, if the correlations can be engineered (through medium properties, geometry, or driving conditions), they may provide an additional degree of freedom for optimizing storage architectures.

The framework’s Phase 2 experimental plan (10×10 storage array) should include systematic measurement of inter-site coupling as a function of spacing, to determine the practical density limit.

6. CLAIMS THAT REMAIN UNVALIDATED

Intellectual honesty requires distinguishing what the Bucher et al. results do and do not support. The following core claims of the TPC framework have no direct experimental corroboration from this study:

6.1 Information Storage in Topological Charge

The framework proposes encoding digital information in the winding number l of stabilized phase singularities, with multiple distinguishable states per site. Bucher et al. tracked singularity dynamics but did not attempt to write, store, or retrieve information in topological charge. The viability of singularity-based information encoding remains an open experimental question.

6.2 Room-Temperature Stability Over Engineering Timescales

The framework requires singularities stable for seconds to years (Chapter 8, Section 8.2). Bucher et al. observed singularity dynamics over femtosecond-to-picosecond timescales. The question of long-term stability in any medium—LC, hBN, or otherwise—is not addressed by their study.

6.3 Photon BEC and Fröhlich Condensation in Engineered Systems

Axiom 2 (Phase Transition Principle) requires a condensation mechanism to stabilize topological structures. The Bucher et al. platform uses phonon polaritons, not photon BEC or Fröhlich condensation. The feasibility of achieving condensation-stabilized singularities in artificial systems remains untested.

6.4 DNA as Existence Proof (Axiom 4)

The biological validation principle—that DNA demonstrates room-temperature photonic storage via the proposed mechanism—is not addressed by Bucher et al. The biophoton coherence question remains open.

6.5 RS2 Dimensional Algebra Mapping (Axiom 3)

The Dimensional Ascent Principle, mapping physical structures to the division algebra hierarchy, is a theoretical framework not testable by the Bucher et al. experimental design. Its validity remains a matter for dedicated investigation.

7. REVISED EXPERIMENTAL PRIORITIES

In light of the Bucher et al. findings, we suggest the following refinements to the TPC framework’s five-phase experimental validation pathway:

Phase 1 (Proof of Concept): Add hBN-based singularity characterization alongside LC experiments. The Bucher et al. methodology (UTEM at fs/nm resolution) provides a benchmark protocol for measuring vortex dynamics in any confining medium.

Phase 2 (Information Encoding): Incorporate inter-singularity correlation measurements. The liquid-like spacing observed by Bucher et al. must be quantified in the target storage medium (LC) to determine achievable array density.

Phase 3 (Coherence Enhancement): No change. Condensation physics remains the critical open question, orthogonal to the Bucher et al. results.

Phase 4 (Dual Channel Prototype): Consider whether hBN or similar polaritonic platforms could serve as a characterization testbed before full LC fabrication.

Phase 5 (System Integration): No change at this stage.

New priority: Characterize the velocity divergence signature near singularity creation/annihilation events in candidate storage media. If this signature is detectable and consistent, it could simplify the write-verification subsystem.

8. CONCLUSION

The publication of Bucher et al. (Nature, March 2026) provides the first direct experimental characterization of optical phase singularity dynamics at resolution sufficient to test foundational claims of the Topological Photonic Computing framework (Vanhorn, January 2026). Four specific points of convergence have been identified:

1. **Topological charge conservation** is directly observed in pair creation/annihilation events, consistent with the framework’s information-integrity mechanism.
2. **Non-particulate character** of singularities is confirmed by superluminal velocities, supporting the Negative Form Principle’s ontological distinction between topology and substance.
3. **Confining media enhance** rather than suppress singularity dynamics, strengthening the framework’s engineering premise.
4. **Ensemble spatial correlations** introduce a new parameter relevant to storage density projections.

These convergences are notable because the two bodies of work were conducted independently, from different theoretical starting points, and for different purposes—one proposing a computing architecture, the other investigating fundamental wave physics. The convergence of conclusions about the same physical objects lends mutual support to both.

At the same time, the framework’s core application claims—information encoding, long-term stability, condensation-mediated storage, and room-temperature operation—remain experimentally unvalidated. The Bucher et al. results corroborate the *substrate physics* but not the *computing architecture*. The distance between “these objects have the right properties” and “these objects can be engineered into a computing system” remains the central open question.

The experimental validation pathway proposed in the original framework remains the appropriate next step. The Bucher et al. findings refine that pathway—suggesting new measurement protocols, alternative substrates, and previously unconsidered design parameters—but do not alter its fundamental structure.

The hole remains the data. The question is no longer whether the hole exists with the right properties, but whether it can be made to hold still long enough to be useful.

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